

Activity 2: Find the Cartesian Product $A \times B \times C$

1. Let $A = \{1, 2, 3, 4, 5\}$

$$B = \{V, W, X, Y, Z\}$$

$$A \times B = \{(1, V), (1, W), (1, X), (1, Y), (1, Z), (2, V), (2, W), (2, X), (2, Y), (2, Z), (3, V), (3, W), (3, X), (3, Y), (3, Z), (4, V), (4, W), (4, X), (4, Y), (4, Z), (5, V), (5, W), (5, X), (5, Y), (5, Z)\}$$

2. Let $A = \{6, 7, 8, 9, 10, 11, 12\}$

$$B = \{A, B, C, D, E, F, G, H, I\}$$

$$A \times B = \{(6, A), (6, B), (6, C), (6, D), (6, E), (6, F), (6, G), (6, H), (6, I), (7, A), (7, B), (7, C), (7, D), (7, E), (7, F), (7, G), (7, H), (7, I), (8, A), (8, B), (8, C), (8, D), (8, E), (8, F), (8, G), (8, H), (8, I), (9, A), (9, B), (9, C), (9, D), (9, E), (9, F), (9, G), (9, H), (9, I), (10, A), (10, B), (10, C), (10, D), (10, E), (10, F), (10, G), (10, H), (10, I), (11, A), (11, B), (11, C), (11, D), (11, E), (11, F), (11, G), (11, H), (11, I), (12, A), (12, B), (12, C), (12, D), (12, E), (12, F), (12, G), (12, H), (12, I)\}$$

3. Let $A = \{1, 2, 3, 4\}$

$$B = \{A, B, C, D\}$$

$$C = \{W, X, Y, Z\}$$

$$A \times B \times C = \{(1, A, W), (1, A, X), (1, A, Y), (1, A, Z), (1, B, W), (1, B, X), (1, B, Y), (1, B, Z), (1, C, W), (1, C, X), (1, C, Y), (1, C, Z), (1, D, W), (1, D, X), (1, D, Y), (1, D, Z), (2, A, W), (2, A, X), (2, A, Y), (2, A, Z), (2, B, W), (2, B, X), (2, B, Y), (2, B, Z), (2, C, W), (2, C, X), (2, C, Y), (2, C, Z), (2, D, W), (2, D, X), (2, D, Y), (2, D, Z), (3, A, W), (3, A, X), (3, A, Y), (3, A, Z), (3, B, W), (3, B, X), (3, B, Y), (3, B, Z), (3, C, W), (3, C, X), (3, C, Y), (3, C, Z), (3, D, W), (3, D, X), (3, D, Y), (3, D, Z), (4, A, W), (4, A, X), (4, A, Y), (4, A, Z), (4, B, W), (4, B, X), (4, B, Y), (4, B, Z), (4, C, W), (4, C, X), (4, C, Y), (4, C, Z), (4, D, W), (4, D, X), (4, D, Y), (4, D, Z)\}$$